Rank-27 BVE Admission and Sweep Bridge Correctness Closure

Tactix engineering handoff

24 August 2026

Executive outcome

The current cumulative Tactix source contains two general repairs. First, BVE now restores Kissat's source-default `eliminateclslim=100` admission on each live irredundant input antecedent, not only on generated resolvents. Second, Substitute preserves Sweep equivalence bridges by counted directed physical entries rather than protecting every binary with the same target pair. The first repair removes a large, objectively worse definition-work tail on immutable Rank 27. The second fixes a DRAT population error exposed by the full Tier-D run; it changes no proof rule or strategy decision.

Exact goldcrest replay now returns UNSAT and ordinary `drat-trim` verifies its 1,031,181,788-byte proof. The complete final-source Tier B plus fixed mechanism-relevant Tier C guard passes all 66 tasks with zero residue; exact-source ASan/UBSan native-LRAT fuzz passes 10,000 cases with zero failures. Final Rank 27 is valid UNSAT in 549.172 seconds, so the BVE mechanism win is accepted but the 300-second engineering objective remains open. No benchmark identity, rank, hash, deadline, elapsed-time arm, per-instance switch, or newly tuned constant enters the solver.

Frozen authority and worse-side localization

The immutable Rank-27 input is `benchmarks/cache/tierD24-26/5911653e4edadec391654c8f06353f9a-oski15a01b06s_opt.cnf.xz`, SHA-256 `5435b1b271b049ed9291ab1b9b297afb33b8d2d3c50ff1b48bfaed4ddf7bb4e9`. Frozen Kissat 4.0.4 proves UNSAT in 278.187 seconds; it was not rerun. A separately pinned current-Kissat diagnostic build takes 286.30 seconds and 16.32 seconds in elimination. The preceding checked Tactix compact-core pin takes 503.787 internal seconds and 94.389 seconds in BVE while following its accepted search trajectory.

Terminal BVE traces named the comparable-work defect. Current Kissat makes 292,771 definition attempts for 18,112,153 embedded ticks. Tactix makes 369,081 attempts for 280,614,670 ticks; 32,932 attempts above 1,024 imported variables alone consume 249,343,427 ticks. Paired campaign `benchmarks/runs/20260824-0332-tierd-rank27-current-bve-396947-occurrence/` finds the first concrete long case at external pivot 396947. Both solvers carry the same $2 + 2$ physical occurrences, choose and eliminate the same predecessor, and leave both pivots unassigned. Kissat then declines 396947 before definition extraction; Tactix imports four clauses spanning 666 variables and spends 674 embedded ticks.

Source inspection resolves the boundary without inference from a benchmark. Kissat's `resolve.c::occurrences_literal` rejects a candidate when any live irredundant antecedent exceeds `eliminateclslim`; the two relevant clauses contain 338 and 403 literals. Tactix had implemented the same named limit only for generated resolvents. The repaired public BVE rule defaults to unrestricted; `DefaultPolicy` installs the source-default value through `DefaultStrategyConfig`,

and BVE observes it before formula mutation, gate discovery, or definition extraction. This keeps proof behavior and strategy ownership separate.

The pre-proof-fix guard already established the performance signal: all 66 candidate/comparator tasks validate with zero residue, candidate/comparator PAR-2 is 1,743.764/2,394.097 seconds (0.728 ratio), mp1 improves from 437.966 to 277.765 seconds with conflicts falling 9.05M to 5.43M, and T105 verifies in 480.267 versus 913.729 seconds. These figures are supporting evidence only; the post-proof-repair source is governed by the final guard below.

DRAT bridge-population defect and repair

The first frozen current-head Tier-D run was canceled after task 21, `goldcrest-and-11`, emitted a 968,410,268-byte DRAT proof rejected at line 13,564,647. Exact campaign `benchmarks/runs/20260824-0337-tierd-goldcrest-proof-reproducer/` reproduces that DRAT stream byte-for-byte while native LRAT independently verifies a 10.207-GB proof. Thus the UNSAT derivation is sound and the defect is DRAT's deletion-by-literal population, not a solver contradiction or checker limitation. All large certificates and decompressed inputs were deleted after checking.

Under the repository's required soundness workflow, authorized Claude Code instrumented the binary lifecycle and localized the defect before its account quota stopped the session. Sweep can create a preserved equivalence bridge whose transformed target pair coincides with an ordinary binary. The proof pass consumes that ordinary copy, but old physical compaction exempted every entry sharing the bridge pair. The kernel therefore retained a clause after the checker population had exhausted all of its copies, and a later lemma using the survivor failed RUP.

The repair counts exact directed entries. An unchanged bridge protects its two physical directions, a bridge with one moved endpoint protects only the stable direction created by the in-place rewrite, and a bridge with both endpoints moved protects none before rebuilding. Ordinary copies that merely share a literal pair follow the normal rewrite/removal path. Regression `test_sweep_bridge_target_pair_collision_consumes_ordinary_copy` checks both kernel multiplicity and DRAT population balance. Release and ASan/UBSan Substitute and Sweep tests pass independently.

Exact repaired-source campaign `benchmarks/runs/20260824-0339-goldcrest-bridge-count-repair/` returns UNSAT and ordinary `drat-trim` verifies the 1,031,181,788-byte proof, SHA-256 `c4f93869031ea1327c585906ebdb6f14c8730b6d4be4cb53af1ca4f478cab513`, in 261.516 seconds. It uses 242,652,058 resolution steps, reports zero RAT lemmas in the core, and leaves no proof, decompressed input, checker product, or scratch residue. Claude's small lifecycle trace is preserved and hashed; its partial proof, input, and throwaway worktree are removed.

Final-source validation

The mandatory exact-source full-width campaign is `benchmarks/runs/20260824-0340-BC-final-bve-bridge-guard/`, array `1645170` and aggregate `1645236`. Each arm validates 17 SAT models and 16 UNSAT proofs with no invalid, regression, missing, cleanup, or residue entry. Candidate PAR-2 is 1,670.643 seconds versus 2,165.228 seconds shipped, a 0.771578 ratio; candidate T105 verifies in 420.837 seconds and candidate mp1 in 260.633 seconds. Summary SHA-256 is `d65b8e67d7fefdbd672a29b7be3f7bf15249e710a8138e635530e4eb1fc4456c`. Every SAT claim is checked against the original CNF, every UNSAT claim is checked by the pinned `drat-trim`, and consumed certificates, checker derivatives, and decompressed inputs are deleted before atomic result publication.

The cumulative Release suite is 73/74. All executable, proof/model, bridge, configuration, event-trace, static rule/policy boundary, and campaign-cleanup tests pass. Only `doc_size` fails on the already recorded living-document debt; repository authority forbids silently deleting the resume and evidence surface to manufacture a green count. `git diff -check` is clean.

Final-source fuzz campaign `benchmarks/runs/20260824-0341-final-bve-bridge-fuzz/`, job `1645237`, runs the frozen ASan/UBSan debug binary under native LRAT while forcing BVE, Substitute, and Sweep with periodic normal scheduling. It completes 10,000 cases with zero failures: 5,188 SAT plus 3,575 UNSAT checked verdicts, 1,237 explicit bounded non-terminations, and 1,458,184 stress applications. BVE fires in 6,161 cases, Substitute in 5,139, and Sweep in 2,753. Twenty-five timeout CNFs are retained as diagnostic inputs; no proof or checker artifact remains. A soundness failure, invalid proof, sanitizer finding, or crash invalidates subsequent performance evidence and re-enters the bounded Claude repair workflow.

Checked Rank 27 and operational handoff

Frozen campaign `benchmarks/runs/20260824-0342-tierd-rank27-final-bve-bridge-witness/` uses a separate read-only solver inode, two CPUs, 16 GiB, a 1,200-second solver limit, and ordinary DRAT verification against the original CNF. It returns UNSAT in 548.645 internal / 549.172 wall seconds with 3,813,669 conflicts, 175,272,633 decisions, and peak solver RSS 646,608 KiB. Its 1,514,395,111-byte proof, SHA-256 `0f99f55ac55571d6f83fe53ec8cb4a6a4c8df0a6e127442e50f47e25bea8bb2d`, verifies in 123.572 seconds and is deleted with all checker scratch.

The intended BVE tail collapses: definition ticks fall from 2,505,837,432 to 115,624,070, definition checks from 1,223,465 to 873,301, and BVE time from 94.389 to 36.947 seconds. Total inprocessing improves 254.377 to 213.590 seconds. The overall target still misses because search/scaffolding rises 249.410 to 335.055 seconds on a changed trajectory. Current Kissat spends 188.64 seconds in search despite more conflicts, decisions, and propagations; its Vivify/Substitute/Congruence/Eliminate times are 26.19/10.42/5.91/16.32 seconds versus Tactix's 66.680/36.659/21.752/36.947. These cross-trajectory totals identify throughput leads, not a same-state authorization to patch any one rule. The next analysis must localize their earliest common-state cost or yield divergence.

After fuzz passes, the same committed Release source is frozen independently for the requested full SC24–26 Tier-D union under 1,800 seconds, 16 GiB, two CPUs per task, and throttle 85. That array uses the accepted baseline harness, validates every model and proof, retains no certificate, and runs asynchronously so it does not block convergence work.

The successor is `benchmarks/runs/20260824-0343-tierd-final-bve-bridge-16g-1800s/`, array `1645240` and dependent aggregate `1645241`. It contains 1,166 tasks, uses an independent read-only solver inode with SHA-256 `5813aca3eb1041ab3d1f398e2f6a17fd3918e3375595aa86a443679bb8445ca9`, and records manifest SHA-256 `82b56c2662b53a1264004cc1f3eed067a57e912e7a1ead1656ec02ba83298cff`. Compute-node SAT/model and UNSAT/DRAT-to-Cake-LPR preflight passes with scratch deleted. The array is active and is not awaited here. An early 48-result audit has zero retained or pending-cleanup artifacts but two resource failures: task 41 (expected SAT) aborts on `std::bad_alloc` at 13,390,560 KiB RSS, and task 74 (baseline UNKNOWN) does likewise at 14,732,416 KiB. Both partial proofs are hashed and deleted. These are solve-loss/memory findings, not accepted solver verdicts; the final aggregate must classify their coverage effect and any later correctness failure before the campaign is used.

The validated source and tests are committed and pushed as `26a2d9c` only after the guard and checked witness establish a coherent boundary; this report and living-document closure follow separately. The wrap-up benchmark hygiene pass removes the 1,330,575-byte superseded, never-

submitted pre-proof-fix Rank-27 scaffold and preserves every active or evidence-bearing campaign. Rank 22 remains parked at its known Vivify boundary; no completed Backbone, Sweep, or Vivify investigation is reopened. If Rank 27 meets 300 seconds, close it and resume the remaining frozen UNSAT ranks from their preserved common-state ledgers. If it misses, keep the accepted BVE admission and bridge repairs and localize the earliest remaining current/current worse-side event; do not undo the exact Kissat effort formula or any prior checked closure.